

1.Docker Installation

```
sudo apt update
sudo apt upgrade -y
sudo apt install -y ca-certificates curl gnupg lsb-release
sudo mkdir -p /etc/apt/keyrings
curl -fsSL https://download.docker.com/linux/ubuntu/gpg | \
sudo gpg --dearmor -o /etc/apt/keyrings/docker.gpg
echo \
"deb [arch=$(dpkg --print-architecture) \
signed-by=/etc/apt/keyrings/docker.gpg] \
https://download.docker.com/linux/ubuntu \
$(lsb_release -cs) stable" | \
sudo tee /etc/apt/sources.list.d/docker.list > /dev/null
sudo apt update
sudo apt install -y docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin
sudo docker --version
sudo docker run hello-world
```

2.Working with Docker

To verify docker version

docker version

```
root@ip-172-31-92-238:~# docker version
Client: Docker Engine - Community
 Version: 29.3.0
  API version: 1.54
  Go version: go1.25.7
  Git commit: 5927d00
  Built: Thu Mar 5 14:25:48 2026
  OS/Arch: linux/amd64
  Context: default

Server: Docker Engine - Community
 Engine:
  Version: 29.3.0
  API version: 1.54 (minimum version 1.40)
  Go version: go1.25.7
  Git commit: 83bca51
  Built: Thu Mar 5 14:25:48 2026
  OS/Arch: linux/amd64
  Experimental: false
 containerd:
  Version: v2.2.2
  GitCommit: 301b2dac98f15c27117da5c9af12118a041a31d9
 runc:
  Version: 1.3.4
  GitCommit: v1.3.4-0-gd6d73eb8
 docker-init:
  Version: 0.19.0
  GitCommit: de40ad0
root@ip-172-31-92-238:~#
```

To pull the image use the below

docker pull python:3.11

```
root@ip-172-31-92-238:~#
root@ip-172-31-92-238:~# docker pull python:3.11
3.11: Pulling from library/python
8cd65a420aac: Download complete
8f6ad858d0a4: Downloading 22.02MB/49.3MB
b012eb15dff0: Downloading 22.02MB/25.62MB
ee3a0e7d77f0: Downloading 25.17MB/67.78MB
66063df90a44: Download complete
8688d0f2f567: Downloading 22.02MB/236.1MB
1589b6e505d3: Downloading 19.92MB/23.98MB
04cce6e61970: Download complete
c9a1b72a890f: Downloading 12.58MB/17.8MB
```

```
66063df90a44: Pull complete
8688d0f2f567: Pull complete
1589b6e505d3: Pull complete
04cce6e61970: Download complete
c9a1b72a890f: Download complete
Digest: sha256:4651409ff55024245f70920259ecce27f2109b0dc0e502a10bce541f1072cd66
Status: Downloaded newer image for python:3.11
docker.io/library/python:3.11
root@ip-172-31-92-238:~#
```

To verify the images in the system

docker images

```
Status: Downloaded newer image for python:3.11
docker.io/library/python:3.11
root@ip-172-31-92-238:~# docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
hello-world:latest	6ae972d1ac53	25.9kB	9.49kB	U
python:3.11	4651409ff550	1.61GB	427MB	

```
root@ip-172-31-92-238:~#
```

To create one container use the below

docker run -itd --name myc1 python:3.11 /bin/bash

```
python:3.11 4651409ff550 1.61GB 427MB
root@ip-172-31-92-238:~# docker run -itd --name myc1 python:3.11 /bin/bash
6e1a13e2ca2195a96a9b0ff28ae7b3899849a3f8afd453aa4edee38939f17f7b
root@ip-172-31-92-238:~#
```

To check the running containers

docker ps

```
6e1a13e2ca2195a96a9b0ff73a57b389d49a318ard453aa4edee38939f1717b
root@ip-172-31-92-238:~# docker ps
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS      NAMES
6e1a13e2ca21   python:3.11  "/bin/bash"             24 seconds ago  Up 23 seconds          myc1
root@ip-172-31-92-238:~#
```

To login to the created container

docker exec -it myc1 /bin/bash

```
See 'docker exec --help' for more information
root@ip-172-31-92-238:~# docker exec -it myc1 /bin/bash
root@6e1a13e2ca21:/#
root@6e1a13e2ca21:/#
root@6e1a13e2ca21:/#
root@6e1a13e2ca21:/#
```

Exit from the container

```
root@6e1a13e2ca21:/#
root@6e1a13e2ca21:/# exit
exit
root@ip-172-31-92-238:~#
```

3. Dockerfile

mkdir /root/app

cd /root/app

```
root@ip-172-31-92-238:~#
root@ip-172-31-92-238:~# mkdir /root/app
root@ip-172-31-92-238:~# cd /root/app/
root@ip-172-31-92-238:~/app# ls -l
```

cat <<EOF > requirements.txt

flask

EOF

cat <<EOF >app.py

```
from flask import Flask, jsonify
app = Flask(__name__)
@app.route("/")
def home():
    return jsonify({"message": "Hello All Welcome to Dvs GenAI Sessions!"})
@app.route("/health")
def health():
    return jsonify({"status": "OK"})

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=5000)
EOF
```

cat <<EOF > Dockerfile

```
FROM python:3.11
EXPOSE 5000
WORKDIR /opt/app
COPY app.py /opt/app
COPY requirements.txt /opt/app
RUN pip install -r requirements.txt
CMD ["python", "app.py"]
EOF
```